Advanced Machine Learning

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University of Bucharest, 2nd semester, 2025-2026

Recap: The fundamental theorem of statistical learning

Theorem (The Fundamental Theorem of Statistical Learning).
Let $\mathcal{H}$ be a hypothesis class of functions from a domain $\mathcal{X}$ to $\{0,1\}$ and let the loss function be the $0-1$ loss. Then, the following are equivalent:

1. $\mathcal{H}$ has the uniform convergence property.
2. Any ERM rule is a successful agnostic PAC learner for $\mathcal{H}$.
3. $\mathcal{H}$ is agnostic PAC learnable.
4. $\mathcal{H}$ is PAC learnable.
5. Any ERM rule is a successful PAC learner for $\mathcal{H}$.
6. $\mathcal{H}$ has a finite VC-dimension.

A finite VC- dimension guarantees learnability. Hence, the VC-dimension characterizes PAC learnability.

Recap: Proof for 6 → 1

We want to prove that finite VC-dimension → *uniform convergence property*

Two steps:

1. (Sauer's lemma) If $\text{VCdim}(\mathcal{H}) \leq d < \infty$, then even though $\mathcal{H}$ might be infinite, when restricting it to a finite set $C \subseteq \mathcal{X}$, its "effective" size, $|\mathcal{H}_C|$, is only $O(|C|^d)$. That is, the size of $\mathcal{H}_C$ grows polynomially rather than exponentially with $|C|$.

2. we have shown in lecture 4 that finite hypothesis classes enjoy the uniform convergence property. We generalize this result and show that uniform convergence holds whenever the hypothesis class has a "small effective size." By "small effective size" we mean classes for which $|\mathcal{H}_C|$ grows polynomially with $|C|$.

Recap: The Growth function

Definition

Let $\mathcal{H}$ be a hypothesis class. Then the growth function of $\mathcal{H}$, denoted by τ_H, where $\tau_{\mathcal{H}}$: **N**→**N**, is defined as:

$$\tau_H(m) = \max_{C \subseteq X : |C| = m} |H_C|$$

In other words, $\tau_H(m)$ is the maximum number of different functions from a set C of size m to {0,1} that can be obtained by restricting $\mathcal{H}$ to C.

Observation: if VCdim($\mathcal{H}$) = d then for any m ≤ d we have $\tau_{\mathcal{H}}(m) = 2^m$. In such cases, $\mathcal{H}$ induces all possible functions from C to {0,1}.

What happens when m becomes larger than the VC-dimension?

Answer given by the Sauer's lemma: the growth function $\tau_{\mathcal{H}}$ increases polynomially rather than exponentially with m.

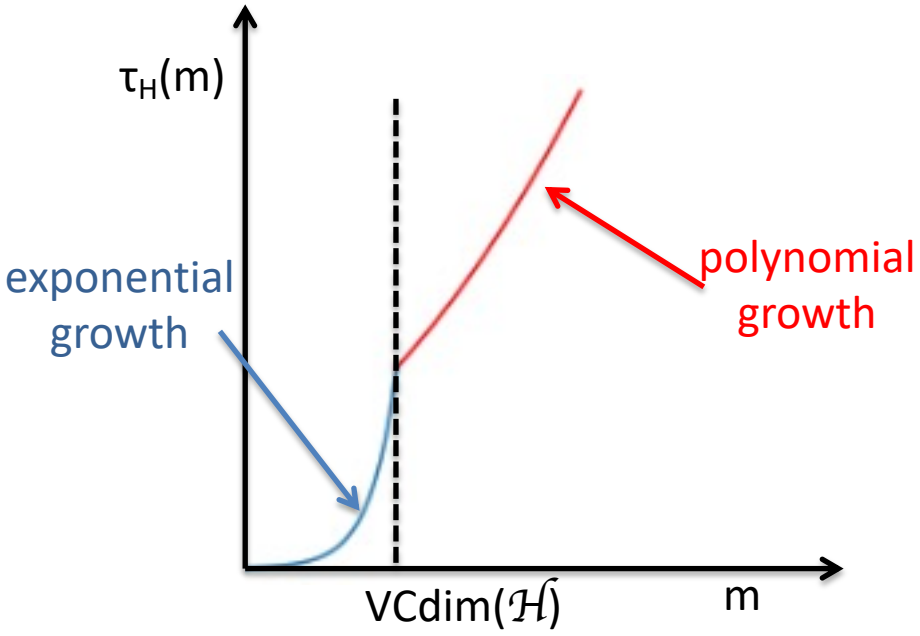
Recap: The Sauer's lemma

Lemma (Sauer – Shelah – Perles)

Let $\mathcal{H}$ be a hypothesis class with $\text{VCdim}(\mathcal{H}) \leq d < \infty$. Then, for all m, we have that:

$$\tau_H(m) \leq \sum_{i=0}^{d} C_m^i$$

In particular, if $m > d+1$ then $\tau_{\mathcal{H}}(m) \leq (em/d)^d = O(m^d)$



Recap: $\tau_{\mathcal{H}}$ grows polynomially

Corollary

Let H be a hypothesis class with VCdim(H) = d. Then for all m ≥ d:

$$\tau_H(m) \leq \left(\frac{em}{d}\right)^d = O(m^d)$$

Proof:

From the Sauer lemma we have:

$$\tau_H(m) \leq \sum_{i=0}^{d} C_m^i \leq \sum_{i=0}^{d} \left(C_m^i \times \left(\frac{m}{d}\right)^{d-i} \right) \leq \sum_{i=0}^{m} \left(C_m^i \times \left(\frac{m}{d}\right)^{d-i} \right) = \left(\frac{m}{d}\right)^d \sum_{i=0}^{m} \left(C_m^i \times \left(\frac{d}{m}\right)^i \right)$$

m ≥ d

m ≥ d

$$\tau_H(m) \leq \left(\frac{m}{d}\right)^d \sum_{i=0}^{m} \left(C_m^i \times \left(\frac{d}{m}\right)^i \right) = \left(\frac{m}{d}\right)^d \left(1 + \frac{d}{m}\right)^m \leq \left(\frac{m}{d}\right)^d \left(e^{\frac{d}{m}}\right)^m = \left(\frac{em}{d}\right)^d$$

Newton's binomial formula

$1-x \leq e^{-x}$

Proof for 6 → 1

We want to prove that finite VC-dimension → *uniform convergence property*

Two steps:

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2. we have shown in lecture 4 that finite hypothesis classes enjoy the uniform convergence property. We generalize this result and show that uniform convergence holds whenever the hypothesis class has a "small effective size." By "small effective size" we mean classes for which $|\mathcal{H}_C|$ grows polynomially with $|C|$.

Uniform converge holds for $\mathcal{H}$ with small effective size

Theorem

Let $\mathcal{H}$ be a class and let $\tau_{\mathcal{H}}$ be its growth function. Then, for every $\mathcal{D}$ and every $\delta \in (0,1)$, with probability of at least $1-\delta$ over the choice of $S \sim \mathcal{D}^m$ we have:

$$\left|L_D(h) - L_S(h)\right| \leq \frac{4 + \sqrt{\log(\tau_H(2m))}}{\delta\sqrt{2m}}$$

Proof:

- in the book, is beyond the scope of this lecture

Proof for 6 → 1

We want to prove that finite VC-dimension → *uniform convergence property.*

Combine the last result with Sauer lemma: $\tau_{\mathcal{H}}(m) \leq (em/d)^d = O(m^d)$ to obtain: for every $\mathcal{D}$ and every $\delta \in (0,1)$, with probability of at least $1-\delta$ over the choice of $S \sim \mathcal{D}^m$ we have:

$$\left|L_D(h) - L_S(h)\right| \leq \frac{4 + \sqrt{\log(\tau_H(2m))}}{\delta\sqrt{2m}} \leq \frac{4 + \sqrt{d\log(2em/d)}}{\delta\sqrt{2m}} \leq \frac{2\sqrt{d\log(2em/d)}}{\delta\sqrt{2m}}$$

Sauer lemma
$\tau_{\mathcal{H}}(2m) \leq (2em/d)^d$

consider m such that
$4^2 \leq d\log(2em/d)$

$$\left|L_D(h) - L_S(h)\right| \leq \frac{1}{\delta}\frac{\sqrt{2d\log(2em/d)}}{\sqrt{m}} < \varepsilon$$

This leads (see the calculation in the book) to:

$$m \geq 4\frac{2d}{(\delta\varepsilon^2)}\log(\frac{2d}{\delta\varepsilon^2}) + \frac{4d\log(2\varepsilon/d)}{(\delta\varepsilon^2)}$$

Proof for 6 → 1

We want to prove that finite VC-dimension → *uniform convergence property.*

for every $\mathcal{D}$ and every $\delta \in (0,1)$, with probability of at least $1-\delta$ over the choice of $S \sim \mathcal{D}^m$ we have that if:

$$m \geq 4\frac{2d}{(\delta\varepsilon^2)}\log(\frac{2d}{\delta\varepsilon^2}) + \frac{4d\log(2\varepsilon/d)}{(\delta\varepsilon^2)}$$

then the sample S is ε-representative

$$\left|L_D(h) - L_S(h)\right| \leq \frac{1}{\delta}\frac{\sqrt{2d\log(2em/d)}}{\sqrt{m}} < \varepsilon$$

So, we have that: $m_H^{UC}(\varepsilon,\delta) \leq 4\frac{2d}{(\delta\varepsilon^2)}\log(\frac{2d}{\delta\varepsilon^2}) + \frac{4d\log(2\varepsilon/d)}{(\delta\varepsilon^2)}$

The derived bound is not the tightest possible, there exist another bound much tighter (see next).

The fundamental theorem of statistical learning – quantitative version

Theorem

Let $\mathcal{H}$ be a hypothesis class of functions from a domain $\mathcal{X}$ to $\{0,1\}$ and let the loss function be the 0−1 loss. Assume that VCdim($\mathcal{H}$) = d < ∞. Then, there are absolute constants C_1, C_2 such that:

1. $\mathcal{H}$ has the uniform convergence property with sample complexity:

$$C_1 \frac{d+\log(1/\delta)}{\epsilon^2} \leq m_{\mathcal{H}}^{UC}(\epsilon,\delta) \leq C_2 \frac{d+\log(1/\delta)}{\epsilon^2}$$

2. $\mathcal{H}$ is agnostic PAC learnable with sample complexity:

$$C_1 \frac{d+\log(1/\delta)}{\epsilon^2} \leq m_{\mathcal{H}}(\epsilon,\delta) \leq C_2 \frac{d+\log(1/\delta)}{\epsilon^2}$$

3. $\mathcal{H}$ is PAC learnable with sample complexity:

$$C_1 \frac{d+\log(1/\delta)}{\epsilon} \leq m_{\mathcal{H}}(\epsilon,\delta) \leq C_2 \frac{d\log(1/\epsilon)+\log(1/\delta)}{\epsilon}$$

The VC dimension determines (along with ε, δ) the samples complexities of learning a class. It gives us a lower and an upper bound.

Intuition for deriving the lower bounds

The PAC case (realizable case)

$$C_1 \frac{d + \log(1/\delta)}{\epsilon} \leq m_{\mathcal{H}}(\epsilon, \delta) \leq C_2 \frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon}$$

Pick a set A = $\{x_1, x_2, \ldots, x_d\}$ of size d (=VCdim($\mathcal{H}$)) that is shattered by $\mathcal{H}$. Choose the following (adversarial) probability distribution $\mathcal{D}$ over $\mathcal{X}$:

$\mathcal{D}(x_1) = 1\text{-}4\varepsilon$, $\mathcal{D}(x_i) = 4\varepsilon/(d\text{-}1)$, i = 2,3,…,d, $\mathcal{D}(x) = 0$, for all x in $\mathcal{X} \setminus A$

By the No Free Lunch theorem as long as a sample S hits B = $\{x_2, \ldots x_d\}$ at most (d-1)/2 times, the probability of making an error over B is ≥ 1/4. This happens because we see less then half of the domain B points. So, our expected error with respect to $\mathcal{D}$ is $4\varepsilon/4 = \varepsilon$.

If the sample S has size *m*, then roughly $4m\varepsilon$ points will hit B = $\{x_2, \ldots, x_d\}$. So, to make less than ε errors we need to have $4m\varepsilon > (d\text{-}1)/2$, $m > (d\text{-}1)/8\varepsilon$

Computational complexity of learning

Computational resources of learning

For learning we need 2 type of resources:

1. *Information = training data*
   - so far we analyzed how much training data (sample size) is required to guarantee, with high probability, that the learned hypothesis achieves an error below a desired threshold.
   - *sample complexity*

2. *Computation = runtime*
   - for how much time an algorithm (that implements learning) will run, once we have sufficiently many training examples
   - *computational complexity*
   - crucial when we need fast ML applications (driver surveillance, stock exchange trading, etc)
   - runtime = number of elementary instructions executed - arithmetic operations over real numbers - in an asymptotic sense (with respect to input size) of the algorithm, e.g. *O(n) – where n is the size of the input size*

Spread Networks

From Wikipedia, the free encyclopedia

Spread Networks is a company founded by Dan Spivey and backed by James L. Barksdale (former CEO of Netscape) that claims to offer Internet connectivity between Chicago and New York City at ultra-low latency (i.e. speeds that are very close to the speed of light), high bandwidth, and high reliability, using dark fiber.[1][2] Its customers are primarily firms engaged in high-frequency trading, where small reductions in latency are important to the extent that they help one close trades before one's competitors.[3][4][5]

History [edit]

The first cable line, running 827 miles (1,331 km) from Chicago (home to the Chicago Mercantile Exchange, where futures and options are traded) to Carteret, New Jersey (home to the Nasdaq data center), laid at a cost of US$300 million, was unveiled in June 2010.[3][5][6] According to a *Forbes* article, the idea for the line first came to Dan Spivey in 2007: Spivey contracted with a New York hedge fund to devise a low-latency arbitrage strategy, wherein the fund would search out tiny discrepancies between futures contracts in Chicago and their underlying equities in New York.[5] Although he successfully created the strategy, he was not able to execute it because he was not able to get access to the market's lowest-latency line.[5] He spent some time researching the feasibility of building an ultra-low-latency line, and then looked for people willing to fund it and found Jim Barksdale.[5] Construction was in full swing (but in extreme secrecy, to avoid getting scooped by competitors) by early 2009.[3][5]

In 2011, Spread Networks expanded to Equinix NY4 IBX data center in Secaucus, New Jersey, 300 Boulevard East in Weehawken, New Jersey, and 165 Halsey Street in Newark, New Jersey.[7][8][9] In October 2012, Spread Networks announced latency improvements, bringing the estimated roundtrip time from 13.1 milliseconds to 12.98 milliseconds.[10] In January 2014, Spread Networks announced that it had opened a point of presence at the NYSE Euronext trading center located in Mahwah, New Jersey.[11]

Announced November 27, 2017, Zayo Group Holdings, Inc. ("Zayo") (NYSE: ZAYO) entered into a definitive agreement to acquire Spread Networks for $127 million in cash.[12][13]

Input size parameter of learning

What should play the role of the input size parameter in learning?

- size of the training set that the algorithm receives?
  - for a very large number of examples, much larger than the sample complexity of learning, the algorithm can ignore the extra samples
  - a larger training set does not make the problem more difficult

- size of the hypothesis class?
  - might be infinite: $|\mathcal{H}_{\text{thresholds}}| = \infty$

- accuracy ε, confidence δ and another parameter n related to the size/complexity of $\mathcal{X}$, $\mathcal{H}$
  - how much computation we need in order to get accuracy ε with confidence δ
  - want to have *efficient learning* (give a formal definition later): polynomial in $1/\varepsilon$, $1/\delta$ and n (some parameter related to the size/complexity of domain/hypothesis class: more complex hypothesis needs more computation time)

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- parameter n can be the embedding dimension
  - if we decide to use n features to describe objects, how will that increase runtime?

- we study the runtime in an asymptotic sense by defining a sequence of pairs $(\mathcal{X}_n, \mathcal{H}_n)_{n=1,2,\ldots}$ and studying asymptotic complexity of learning $\mathcal{X}_n$, $\mathcal{H}_n$ as n grows to ∞

Prevent "cheating"

The output of the learning algorithm L is a hypothesis h from $\mathcal{H}$.

- a learning algorithm L can "cheat" by transferring the computational burden to the output hypothesis
  - define the output hypothesis to be the function that stores the training set in memory and computes the ERM hypothesis on the training set and applies it to a test example x

- the runtime of a learning algorithm A defined as the maximum of:
  - the time it takes A to output some h
  - the time it takes h to output a label on any given x from $\mathcal{X}$

Example 1: Conjunctions of Boolean literals

- $\mathcal{H}_{\text{conj}}^{\text{d}}$ = class of conjunctions of at most d Boolean literals $x_1, \ldots, x_{\text{d}}$
  - a Boolean literal is either x_{i} or its negation $\overline{x_i}$ (or 1 = missing literal)
  - can interpret x_{i} as feature i
  - example: $h = x_1 \wedge \overline{x_2} \wedge x_4$ where $\overline{x_2}$ denotes the negation of the Boolean literal x_2
  - $\mathcal{X} = \{0,1\}^{\text{d}}$
- consider the realizable case
  - there is a conjunction h^* in $\mathcal{H}_{\text{conj}}^{\text{d}}$ that labels the examples

- $|\mathcal{H}_{\text{conj}}^{\text{d}}| = 3^{\text{d}} + 1 < \infty$ so it has finite VC dimension (less than $\log_2(3^{\text{d}}+1)$), so it's PAC learnable. In seminar class 3 we shown that $\text{VCdim}(\mathcal{H}_{\text{conj}}^{\text{d}}) = \text{d}$ so the sample complexity $m_{\mathcal{H}}(\varepsilon,\delta)$ is bounded by:

$$C_1 \frac{d + \log(1/\delta)}{\epsilon} \leq m_{\mathcal{H}}(\epsilon,\delta) \leq C_2 \frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon}$$

- So, $m_{\mathcal{H}}(\varepsilon,\delta)$ is polynomial in $1/\varepsilon$, $1/\delta$, d (measures the complexity of the hypothesis class $\mathcal{H}_{\text{conj}}^{\text{d}}$)

Example 1: Conjunctions of Boolean literals

- $\mathcal{H}_{\text{conj}}^{d}$ = class of conjunctions of at most d Boolean literals $x_1, \ldots, x_d$
- a simple algorithm for finding an ERM hypothesis is based on positive examples and consists of the following:
  - for each positive example $(b_1, \ldots b_d)$,
    - if $b_i = 1$ then $\overline{x_i}$ is ruled out as a possible literal in the concept class
    - if $b_i = 0$ then x_i is ruled out.
  - the conjunction of all the literals not ruled out is thus a hypothesis consistent with the target

0	1	1	0	1	1	+
0	1	1	1	1	1	+
0	0	1	1	0	1	-
0	1	1	1	1	1	+
1	0	0	1	1	0	-
0	1	0	0	1	1	+

$x_1 \; \overline{x_1} \; x_2 \; \overline{x_2} \; x_3 \; \overline{x_3} \; x_4 \; \overline{x_4} \; x_5 \; \overline{x_5} \; x_6 \; \overline{x_6}$

$\rightarrow \overline{x}_1 \wedge x_2 \wedge x_5 \wedge x_6$

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    - if $b_i = 0$ then x_i is ruled out.
  - the conjunction of all the literals not ruled out is thus a hypothesis consistent with the target

- runtime of the algorithm is $O(m_{\mathcal{H}}(\varepsilon,\delta)*\text{d})$, so is polynomial in $1/\varepsilon$, $1/\delta$, d

- in the agnostic (unrealizable) case: unless P = NP, there is no algorithm whose running time is polynomial in $m_{\mathcal{H}}(\varepsilon,\delta)$ and d that is guaranteed to find an ERM hypothesis for the class of Boolean conjunctions.

- **from the statistical perspective of learning, there is no difference between the realizable and agnostic cases (a class is learnable in both cases if and only if it has a finite VC-dimension). In contrast, from the computational perspective of learning the difference is immense.**

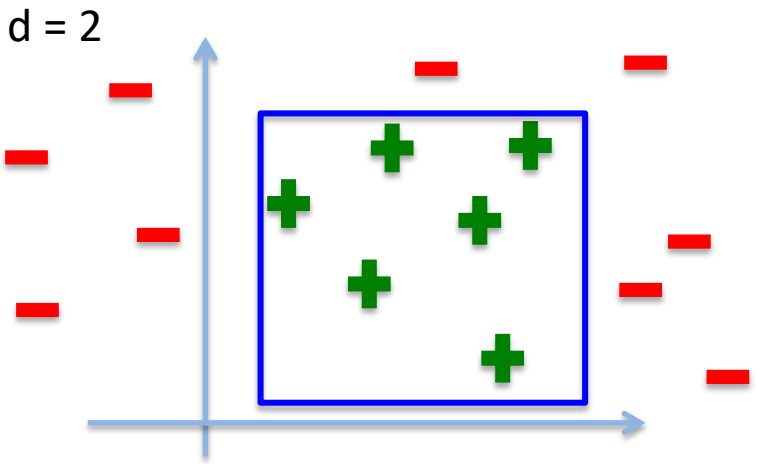
Example 2: Learning axis aligned rectangles in $\mathbf{R}^d$

- $\mathcal{H}_{rec}^d$ = the class of axis aligned rectangles in $\mathbf{R}^d$

$$H_{rec}^d = \{h_{a_1,b_1,a_2,b_2,\ldots,a_d,b_d} : R^d \to \{0,1\} \mid a_1 \le b_1, a_2 \le b_2, \ldots, a_d \le b_d, a_i \in R, b_i \in R\}$$

$$h_{a_1,b_1,a_2,b_2,\ldots,a_d,b_d}(x_1, x_2 \ldots, x_d) = \begin{cases} 1, if\ a_1 \le x_1 \le b_1, a_2 \le x_2 \le b_2, \ldots, a_d \le x_d \le b_d \\ 0, otherwise \end{cases}$$

- consider the realizable case:
  - there exists a rectangle h* in $\mathcal{H}_{rec}^d$ with real risk = 0



We have shown in the seminar class that:

- the algorithm that returns the rectangle enclosing all positive examples is ERM
- $\mathcal{H}_{rec}^d$ is PAC learnable with sample size

$$m_{H_{rec}^d}(\varepsilon, \delta) \le \left\lceil \frac{2d \log(\frac{2d}{\delta})}{\varepsilon} \right\rceil$$

- the runtime is $O(m_H d)$ as for each dimension, the algorithm has to find the minimal and the maximal values among the positive instances in the training sequence. So it is polynomial in $1/\varepsilon$, $1/\delta$, d.

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- consider the agnostic case:
  - distribution $\mathcal{D}$ over $\mathcal{Z} = \mathbf{R}^d \times \{0,1\}$ (a sample could get both labels)
  - if there exist a labeling function f this might not be in $\mathcal{H}_{\text{rec}}^{d}$

- VCdim($\mathcal{H}_{\text{rec}}^{d}$) = 2d (see seminar class), so we have that:

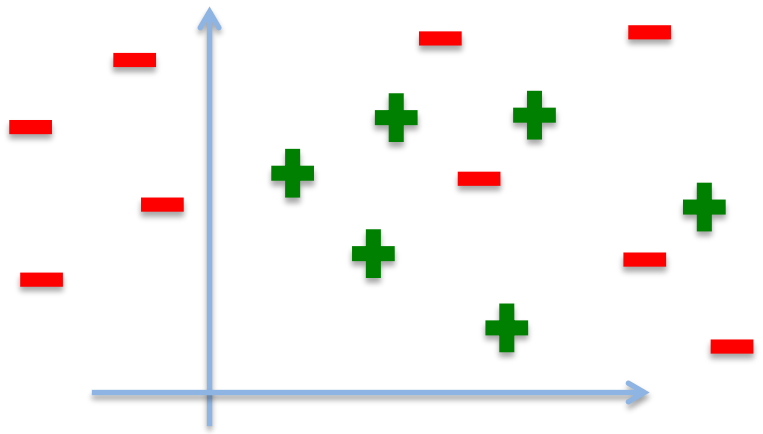
$$C_1 \frac{2d + \log(\frac{1}{\delta})}{\varepsilon^2} \le m_{H_{rec}^{d}}(\varepsilon, \delta) \le C_2 \frac{2d + \log(\frac{1}{\delta})}{\varepsilon^2}$$

- $m_{H_{rec}^{d}}(\varepsilon, \delta)$ is polynomial in 1/ε, 1/δ , d (measures the complexity of the $\mathcal{H}_{\text{rec}}^{d}$)

Example 2: Learning axis aligned rectangles in $\mathbf{R}^d$

- $\mathcal{H}_{rec}^{d}$ = the class of axis aligned rectangles
- consider that we have a sample S of size: $m_{H_{rec}^{d}}(\varepsilon,\delta) \approx C\frac{2d+\log(\frac{1}{\delta})}{\varepsilon^2}$
- what is the runtime of the ERM algorithm?
  - how long it will take to find the best rectangle in $\mathbf{R}^d$?
  - go over all axis aligned rectangles in $\mathbf{R}^d$ and choose the best one (based on minimizing the error on the training data)

d = 2



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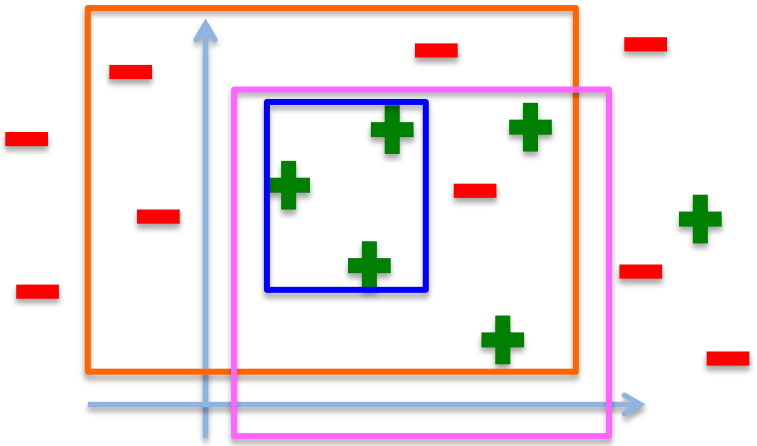

error: (1 + 4)/ (6 + 9) = 5/15

error: (3 + 0)/ (6 + 9) = 3/15

error: (1 + 1)/ (6 + 9) = 2/15

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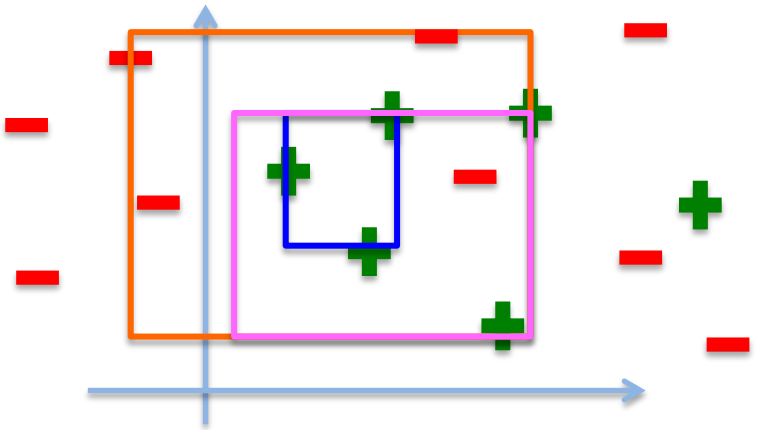

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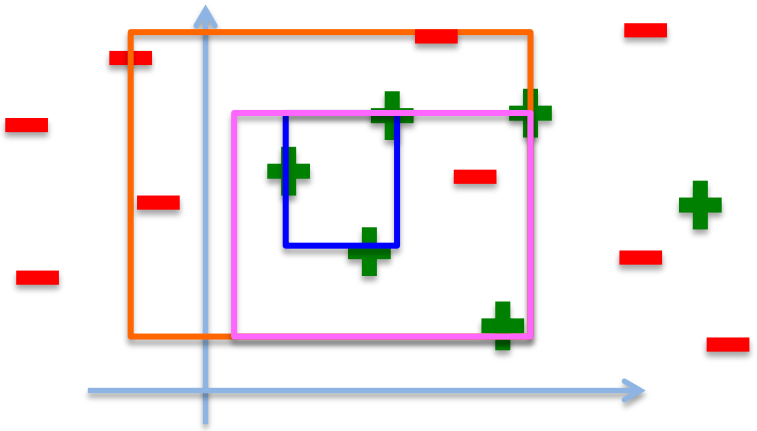
- the number of all possible rectangles can be reduces to all possible rectangles that have points of S on every boundary edge (very efficient algorithm)

Example 2: Learning axis aligned rectangles in $\mathbf{R}^d$

- $\mathcal{H}_{rec}^{d}$ = the class of axis aligned rectangles
- consider that we have a sample S of size: $m_{H_{rec}^{d}}(\varepsilon,\delta) \approx C\frac{2d+\log(\frac{1}{\delta})}{\varepsilon^2}$
- what is the runtime of the ERM algorithm?
  - how long it will take to find the best rectangle in $\mathbf{R}^d$?
  - Step 1: generate all the rectangles based on the sample points in $\mathbf{R}^d$
  - Step 2: for each such rectangle compute the training error
  - Step 3: choose the rectangle with the smallest training error



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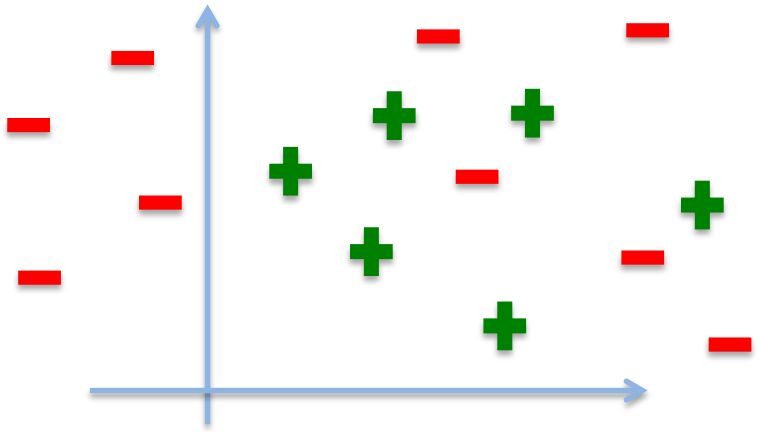
- how many possible rectangles can we construct based on the points in the sample S?

Example 2: Learning axis aligned rectangles in $\mathbf{R}^d$

- $\mathcal{H}_{rec}^{d}$ = the class of axis aligned rectangles in $\mathbf{R}^d$
- how many possible rectangles can we construct based on the points in the sample S?

d = 1

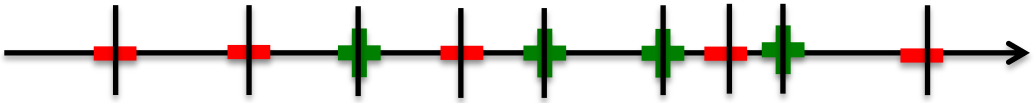
d = 2



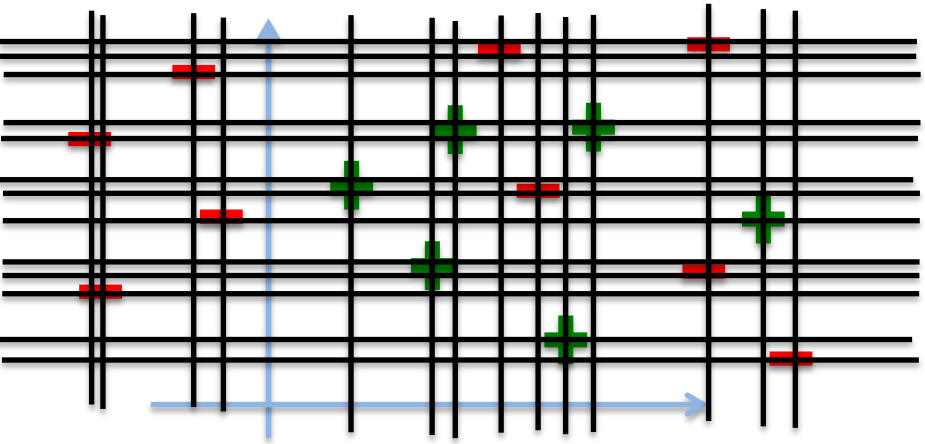
Example 2: Learning axis aligned rectangles in $\mathbf{R}^d$

- $\mathcal{H}_{rec}^{d}$ = the class of axis aligned rectangles in $\mathbf{R}^d$
- how many possible rectangles can we construct based on the points in the sample S?

d = 1



d = 2



Every such rectangle is determined by at most 2d points from S

So there are at most $|S|^{2d}$ such rectangles.

For each rectangle we need to iterate over all examples to compute the training error.

So, the runtime is: $O\left(\left[C\frac{2d+\log(\frac{1}{\delta})}{\varepsilon^2}\right]^{2d+1}\right)$

Example 2: Learning axis aligned rectangles in $\mathbf{R}^d$

- $\mathcal{H}_{rec}^{d}$ = the class of axis aligned rectangles in $\mathbf{R}^d$

- the runtime of the ERM_H is: $O\left(\left\lceil C\frac{2d+\log(\frac{1}{\delta})}{\varepsilon^2}\right\rceil^{2d+1}\right)$

- for every fixed dimension *d,* ERM_H can be implemented in time which is polynomial in $1/\varepsilon$, $1/\delta$, *d* (measures the complexity of the $\mathcal{H}_{rec}^{d}$) therefore we have efficient learning (see the formal definition later)

- however, as a function of d the runtime of the algorithm implementing the ERM_H presented is exponential in *d.* It can be proved that there is no better algorithm (unless P = NP) than the one proposed.

- **statistical vs computational perspective of learning**

Formal definition of efficient learning

Definition

Given a function $f : (0,1)^2 \rightarrow \mathrm{N}$, a learning task $(\mathcal{Z}, \mathcal{H}, \ell)$, and a learning algorithm A, we say that A solves the learning task in time $O(f)$ if there exists some constant number c, such that for every probability distribution $\mathcal{D}$ over $\mathcal{Z}$, and input $\varepsilon, \delta \in (0,1)$, when A has access to samples generated i.i.d by $\mathcal{D}$, we have that:

- A terminates after performing at most $c * f(\varepsilon, \delta)$ operations;
- the output of A, denoted h_A, can be applied to predict the label of a new example while performing at most $c * f(\varepsilon, \delta)$ operations;
- the output of A is probably approximately correct; namely, with probability of at least $1 - \delta$ (over the random samples A receives):

$$L_{\mathcal{D}}(h_A) \leq \min_h L_{\mathcal{D}}(h) + \varepsilon$$

Formal definition of efficient PAC learning (Valiant 1984)

In 1984, Leslie Valiant defined efficient PAC learning: PAC learnability + require the number of examples and the runtime of the algorithm A (training + testing) to be polynomial in *1/ε, 1/δ, n*.

Example:

- $\mathcal{U}_n = \{h: B^n \rightarrow \{0,1\}\}$ - the concept class formed by all subsets of B^n
- $|\mathcal{U}_n| = 2^{2^n}$ - finite, so is PAC learnable with $m_{\mathcal{H}}(\varepsilon, \delta)$ in the order of m:

$$m \geq \left\lceil \frac{1}{\varepsilon}\left(2^n \log(2) + \log(\frac{1}{\delta})\right)\right\rceil$$

- $VCdim(\mathcal{U}_n) = 2^n$ is not polynomial in n
- sample complexity exponential in *n*, number of variables
- it is not efficient PAC-learnable in any practical sense (need polynomial sample complexity)

Relation between consistency and PAC learning

- study the runtime of implementing the ERM rule in the PAC scenario
- a learning rule is consistent if:
  - input: $\mathcal{H}$ and $S = (x_1, y_1), \ldots, (x_m, y_m)$
  - output: $h \in \mathcal{H}$, h is an ERM hypothesis, i.e. $h(x_i) = y_i$
- if VCdim($\mathcal{H}$) $\leq$ d then $\mathcal{H}$ is PAC learnable (in the realizable case) by the consistent rule with sample complexity (given by the fundamental theorem):

$$C_1 \frac{d + \log(1/\delta)}{\epsilon} \leq m_{\mathcal{H}}(\epsilon, \delta) \leq C_2 \frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon}$$

- if d is polynomial in n and the consistent rule has runtime polynomial in the sample size $m_{\mathcal{H}}(\varepsilon, \delta)$ then we have efficient PAC learning
- so, efficient 'consistent-hypothesis-finder' $\rightarrow$ efficient PAC learning
- does the converse implication holds?
  - yes, based on randomized algorithms (next lecture)
- we will show that there exist classes that are PAC learnable (in the realizable case) for which the sample complexity is polynomial but the runtime is not polynomial.